

Assignment nr. 4 – Patch antenna

Assigned: Friday April 17th, 2026
Due: Wednesday May 6th, 2026

Notes

To take the exam with the grading of submitted assignments + oral discussion it is necessary:

- to solve **individually** the exercises;
- to submit the solution **within the deadline**, uploading the PDF in the section “Elaborati” of the web portal

Problem nr. 1

Design a rectangular patch antenna at the (resonance) frequency $f_0 = 5.8$ GHz with $Z_{in}(f_0) = R_0 \leq 120 \Omega$ input impedance, i.e. find the necessary length L and width W . The substrate is an FR4 material, with $\epsilon_r = 4.45$, $h = 1.55$ mm. Use of the provided script for Grad is recommended.

Problem nr. 2

To have an idea of the bandwidth of the antenna designed in Problem 1, calculate the input impedance in the band $\pm 5\%$ around the center frequency f_0 ; use the TL model and consider a constant $G_{edge} = 1/2 G_{rad}(f_0)$ in the whole band:

1) plot real and imaginary parts of $Z_{in}(f)$ on the same graph;

2) calculate and plot the frequency behavior of $|\Gamma_{in}(f)|$, where

$$\Gamma_{in}(f) = \frac{(Z_{in}(f) - Z_0)}{(Z_{in}(f) + Z_0)}$$

and $Z_0 = R_0 = 120\Omega$ (the input impedance at resonance) in the same band.

What is the absolute (MHz) and relative (fractional) ($\pm \%$) bandwidth for $VSWR \leq 2$?

3) In order to appreciate the difference from the approximation $L \approx 0.5 \frac{\lambda_0}{\sqrt{\epsilon_r}}$ and the value you found, repeat

1) adding (on the same plot) the response for the same W but for $L = 0.5 \frac{\lambda_0}{\sqrt{\epsilon_r}}$ (note ϵ_r and not $\epsilon_{r,eff}$)

and without the excess length ΔL .

Problem nr. 3

Referring to the patch antenna designed in Problem 1, find the (approximate) position of the probe to get a 50Ω input resistance.

